

Important questions from previous papers

BLOCK CHAIN

Module1

1. Explain the growth of block chain with network view block diagram.
2. With a neat diagram, explain the generic elements of block chain
3. Define Block chain. Explain the different types of block chain.
4. Explain the following: a) CAP Theorem b) Zero-knowledge proof system
5. What is distributed systems? Explain with neat diagram.
6. What are the benefits and limitations of block chain
7. Why is distributed system essential for block chain technology? Explain.
8. What is consensus? Explain its mechanism. Briefly write about some of the consensus algorithms.

Module2

1. Explain decentralization in blockchain technology, What are the different methods of decentralization.
2. Briefly explain the taxonomy of the cryptographic primitives
3. What is public key cryptography? Explain the working of asymmetric cryptography
4. Explain the following: a) Hash function b) Digital Signature
5. Explain decentralization using blockchain with a neat diagram for different types of network
6. Explain the methods of Decentralization.
7. Define cryptography. Explain the cryptographic primitives with a neat diagram
8. Explain the role of public and private key with respect to Asymmetric cryptography with a neat diagram

Module3

1. What is Bitcoin? Explain its Emergence
2. Write a short notes on : a) Bitcoin limitation b) Litecoin
3. Explain transactions and its life cycle.
4. Write short notes on : a) Proof-of-work b) Proof-of-stake
5. What is Bitcoin? Explain public private address in Bitcoin
6. Explain about transactions life cycle
7. What is Namecoin? Explain merged mining with diagram
8. Write a short note on: a) Prime coin b) Zcash
9. Explain bitcoin scripting languages and their uses

10. Write a brief on hybrid model of bitcoin
11. Write a note on the following a) GRAY model b)RLA model
12. Write a short note on : a)POW as random oracle b)Bitcoin limitations c)Privacy and Anonymity

Module4

1. What is Ethereum?Explain the components of ethereum eco-system with neat diagram
2. What is a smart contract?Give some use cases
3. Specify the properties of smart contract
4. Explain in detail the structure of Ethereum virtual machine
5. What is hyperleader? Discuss on the hyper leader projects
6. Discuss about how smart contracts can be deployed on a block chain
7. Discuss Ricardian Contracts indetail with their properties
8. Explain the Ethereum block chain structure
9. Write a short notes on : a)Wallets b)Smart contracts

Module5

1. How does blockchain technology benefit. The health and finance industry.
2. Write a short notes on: a)IOT blockchain b)e-governance
3. Give an example of how a government is actively promoting the use of block chain
4. Mention some of the alternative block chain and explain any one in detail
5. Explain new Blockchain named solutions like Kadema and Ripple
6. What is Interledge protocol? Explain 4 layers belonging to it.
7. Explain the usage blockchain in Internet of things(IOT).
8. List out and explain about the various applications of blockchain in E-government
9. List out the limitations and applications of the blockchain technology
10. What are the Myths vs reality of block chain technology
11. Explain the following: a)Cryptocurrency b)Hyperledges fabric
12. Write a note on : a) E-Governance b)Contract enforcement mechanism